Introduction/Design Objective:

- The design explores different parameters of Mach-Zehnder Interferometer (MZI) and how they impact the waveguide Free Spectral Range (FSR) and Transmission spectrum.
- The design parameters include path length difference (Δ L), waveguide width (w), incident light polarization (TE, TM) and different bending radii.
- The objective is also to understand the gap between design and fabrication, their origin and impact.

Table 1. Design parameter overview

No	Width (nm)	Δ L (um)	Input pol	Bending radius (um)
1	500	105.096	TE	5
2	500	36.057		5
3	500	23.486		5
4	500	44.002	TM	5

Theory:

- The principal building block of the MZI is a strip waveguide (width = 500 nm, height = 220 nm), grating couplers and Y branch.
- There are in total of 4 design variations, as listed in Table 1.
- Variation on the waveguide width and polarization influences the effective mode index, group index as well as the electric field.
- Variation on the path length difference (Δ L) impacts the FSR.

Modeling and Simulation:

- Table 2 shows the simulated effective index, group index, FSR for all 10 designs. The result is obtained from Tidy 3D simulation at 1550 nm.

Table 2. Effective index, group index and FSR for designs

No	Width (nm)	Δ L (um)	Input pol	Bending radius (um)	n_{eff}	n_g	FSR (nm)
1	500	105.096	TE	5	2.512	4.205	5.43
2	500	36.057		5	2.512	4.205	15.8
3	500	23.486		5	2.512	4.205	24.32
4	500	44.002	TM	10	1.872	4.206	12.98

- Figure 1a, 1b, 1c shows the Ey field for the TE polarization WG with strip width of 500 nm, which corresponds to design 1, 2, 3.

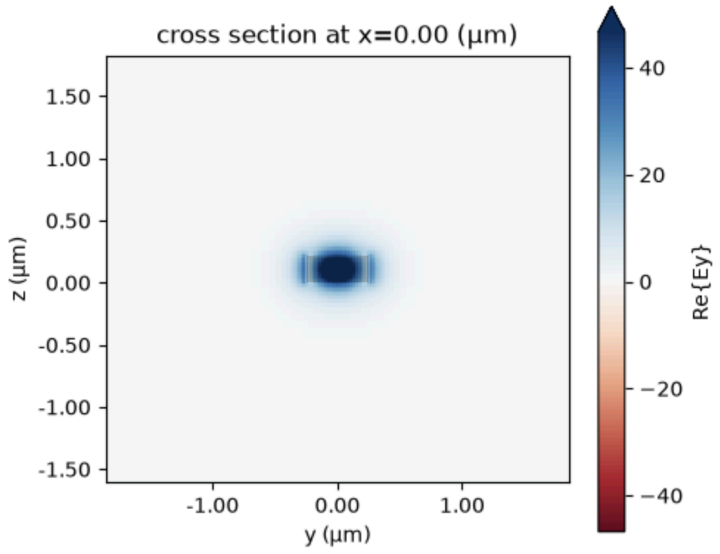


Fig 1a. TE pol, width = 500nm, Ey, 1550 nm

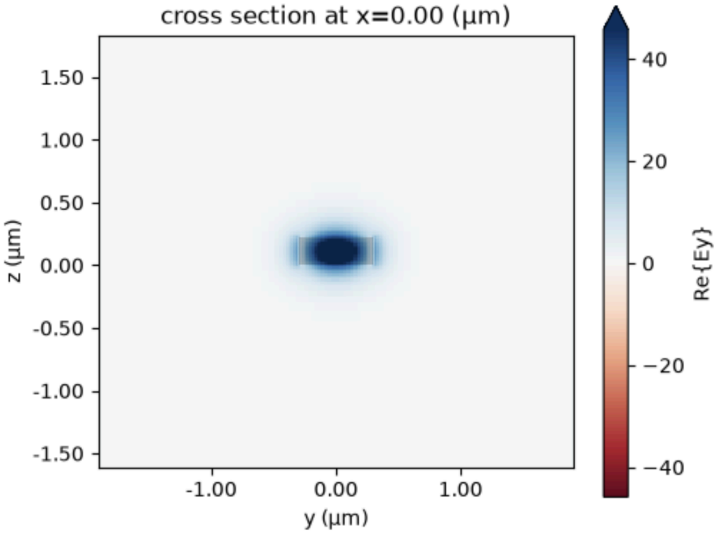


Fig 1b. TE pol, width = 600nm, Ey, 1550 nm

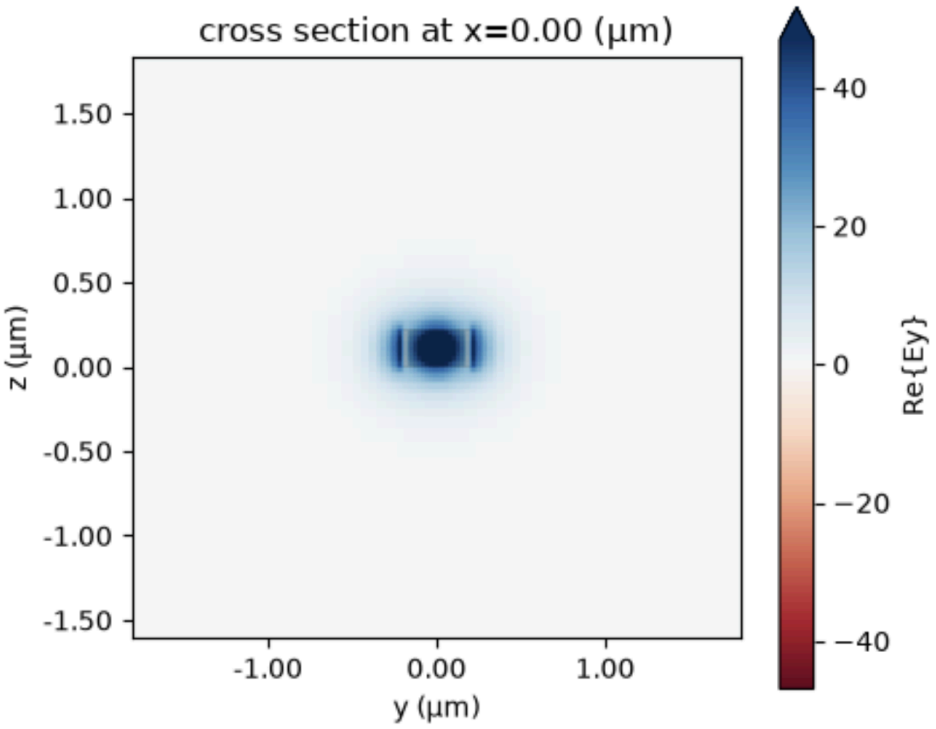


Fig 1c. TE pol, width = 400nm, Ey, 1550 nm

- Figure 2 shows the Ez field of TM pol waveguide with 500 nm width.

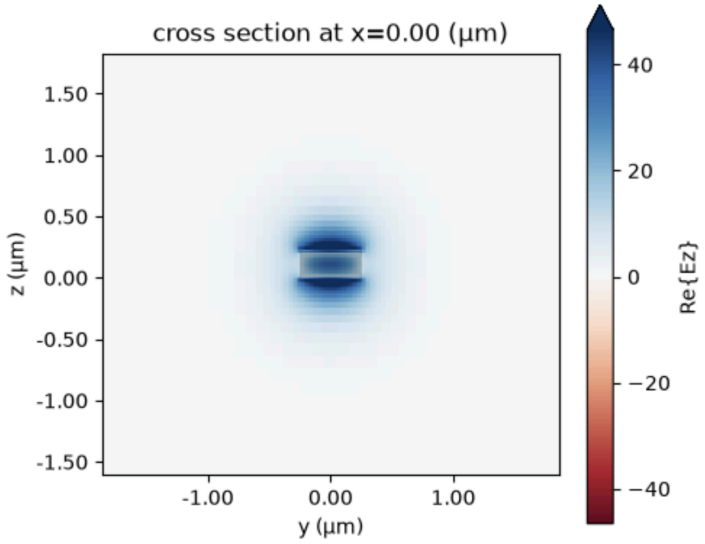


Fig 2. TM pol, width = 500nm, Ey, 1550 nm

- Compact model for the 500 nm width waveguide at 1550 nm, for TE polarization. Simulated in MEEP.
  $n_g(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2$

  $n_g(\lambda) = 2.43 - 1.05(\lambda - 1.55) - 0.134(\lambda - 1.55)^2$

- Transfer function of the interferometer versus wavelength

  $$I_o = I_i/2[1 + \cos(\Delta\beta L)$$

- Transmission spectrum of the MZI with strip waveguide of 500nm width, path delay of 25 um, TE mode, 1550 nm, with assumptions of lossless couplers and y branch, no loss at waveguide bending

